

Radioactive Decay

NE 630

Lesson 03

Prepare: Read FNRP Sections 1.6–1.7 and define the key terms before class.

Macro objective

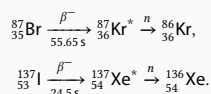
Determine the amounts of a radioactive species and its daughters at any point in time from initial inventories, production rates, and decay constants.

Vocabulary: radioactivity; λ ; half-life; mean lifetime; activity; Bq; Ci; initial-value problem; saturation; steady state; secular equilibrium

1 Why radioactive species appear

Fission-product bridge from Lesson 2

Fission fragments are neutron rich. Their β^- decays produce delayed β , $\bar{\nu}$, and γ energy. When a daughter state is formed above its neutron-separation energy, one decay branch can emit a delayed neutron:



Delayed neutron energies are typically
Deposited β and γ energy drives spent-fuel decay heat; antineutrino energy escapes.

2 The single-nuclide decay law

From a short-time probability to $N(t)$

For $N(t)$ radioactive nuclei,

$$\Delta N \approx -\lambda N(t) \Delta t \implies \boxed{N' = -\lambda N}, \quad N(0) = N_0.$$

Separate and integrate:

$$\frac{dN}{N} = -\lambda dt, \quad \ln \frac{N(t)}{N_0} = \dots\dots\dots,$$

$$\boxed{N(t) = \dots\dots\dots}$$

Here λ is the decay probability per unit time for one nucleus.

Half-life check

$$\frac{N(nt_{1/2})}{N_0} = \dots\dots\dots, \quad \frac{A(nt_{1/2})}{A_0} = \dots\dots\dots$$

What assumptions make both ratios equal to 2^{-n} ?

3 Clocks, number, mass, and activity

Fixed relations and unit conversions

$$\lambda = \frac{\ln 2}{t_{1/2}} = \frac{1}{\bar{t}}, \quad \boxed{A(t) = \lambda N(t)}.$$

number-mass $N = (m/M)N_A$
activity decays per unit time
Bq $1 \text{ Bq} = 1 \text{ s}^{-1}$
Ci $1 \text{ Ci} = 3.7 \times 10^{10} \text{ Bq}$

When there is no production, $A(t) = -dN/dt$. Thus, for a pure radionuclide,

$$m \xrightarrow{\div M} \text{mol} \xrightarrow{\times N_A} N \xrightarrow{\times \lambda} A.$$

Reverse the conversion route

Express the radionuclide mass in terms of measured activity:

$$m = \dots\dots\dots$$

List the units you would use for each quantity.

4 Production and saturation

Decay with a source and saturation

If nuclei are added at rate $R(t)$,

$$\boxed{N' = -\lambda N + R(t)}, \quad N(t_0) \text{ given.}$$

The integrating factor $e^{\lambda t}$ gives

$$\frac{d}{dt}(Ne^{\lambda t}) = R(t)e^{\lambda t},$$

$$\boxed{N(t) = N(t_0)e^{-\lambda(t-t_0)} + \int_{t_0}^t R(t')e^{-\lambda(t-t')} dt'}.$$

For $R(t) = R_0$ and $t_0 = 0$,

$$\boxed{N(t) = N_0 e^{-\lambda t} + \frac{R_0}{\lambda} (1 - e^{-\lambda t})}.$$

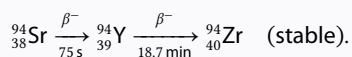
For $N_0 = 0$,

$$A(t) = \dots\dots\dots, \quad N_{\text{sat}} = \dots\dots\dots, \quad A_{\text{sat}} = \dots\dots\dots$$

After n half-lives, $A/A_{\text{sat}} = 1 - 2^{-n}$; one, three, five, and seven half-lives give 50%, 87.5%, 96.9%, and 99.2% of saturation.

5 The $A = 94$ fission-product chain

Chain and initial data from the notes



Assume $N_{\text{Sr}}(0) = N_0$, $N_{\text{Y}}(0) = N_{\text{Zr}}(0) = 0$, and no production after $t = 0$.

Balance each nuclide

$$N'_{\text{Sr}} = -\lambda_{\text{Sr}} N_{\text{Sr}},$$

$$N'_{\text{Y}} = \lambda_{\text{Sr}} N_{\text{Sr}} - \lambda_{\text{Y}} N_{\text{Y}},$$

$$N'_{\text{Zr}} = \lambda_{\text{Y}} N_{\text{Y}}.$$

Label each term as a gain or loss. Why can this one-way chain be solved from the parent downstream?

Sequential solution

$$N_{\text{Sr}}(t) = N_0 e^{-\lambda_{\text{Sr}} t},$$

$$N_{\text{Y}}(t) = \frac{\lambda_{\text{Sr}} N_0}{\lambda_{\text{Y}} - \lambda_{\text{Sr}}} (e^{-\lambda_{\text{Sr}} t} - e^{-\lambda_{\text{Y}} t}),$$

$$\begin{aligned} N_{\text{Zr}}(t) &= \int_0^t \lambda_{\text{Y}} N_{\text{Y}}(t') dt' \\ &= N_0 - N_{\text{Sr}}(t) - N_{\text{Y}}(t). \end{aligned}$$

The last line also follows from conservation of the $A = 94$ nuclei.

Inventory after one hour

At $t = 1$ h,

$$\frac{N_{\text{Sr}}}{N_0} = \text{_____}, \quad \frac{N_{\text{Y}}}{N_0} = \text{_____}, \quad \frac{N_{\text{Zr}}}{N_0} = \text{_____}.$$

Which half-life controls the long-time approach to stable zirconium?

6 General chains and equilibrium

One-way chain with production

For species $i = 0, 1, \dots$,

$$N'_i = -\lambda_i N_i + \lambda_{i-1} N_{i-1} + R_i(t), \quad \lambda_{-1} N_{-1} \equiv 0.$$

Read each equation as

rate of change = parent gain + external gain – decay loss.

Initial values $N_i(0)$ complete the system. Solve the first equation, insert it into the second, and continue downstream. Branching ratios multiply the corresponding parent-gain terms.

FNRP extension: secular equilibrium

For a parent 1 and daughter 2 with $N_1(0) = N_{1,0}$, $N_2(0) = 0$, no direct sources, and a 100% parent-to-daughter branch,

$$N_2(t) = \frac{\lambda_1 N_{1,0}}{\lambda_2 - \lambda_1} (e^{-\lambda_1 t} - e^{-\lambda_2 t}).$$

If $\lambda_1 \ll \lambda_2$, then after several daughter half-lives but before the parent changes appreciably,

$$A_2(t) \approx \text{_____}, \quad \frac{N_2}{N_1} \approx \text{_____}.$$

Explain why equal activities do not imply equal inventories.

Steady state versus secular equilibrium

Steady state: production balances removal for one species, so $N' = 0$ and its inventory is constant.

Secular equilibrium: a short-lived daughter tracks a much longer-lived parent, so their activities are approximately equal while both may vary slowly.

Choose the model before solving

no source, one species	$N' = -\lambda N$
source, one species	$N' = -\lambda N + R(t)$
one-way chain	coupled gain–loss equations
long-lived parent	test for secular equilibrium